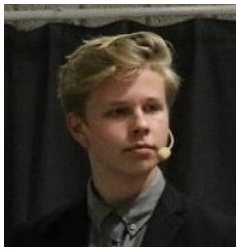

10 November 2024

Felix Steinberger Eriksson
Peterhouse
CB2 1RD, Cambridge, UK
Phone no.: +4676 830 63 34
E-mail: felix.s.eriksson@gmail.com

Da Vinci

Hello,

I recently started my Master's degree in Applied Mathematics at the University of Cambridge after completing my Bachelor's degree in Engineering Mathematics at KTH Royal Institute of Technology in Stockholm, Sweden, also having spent a semester last year studying Mathematics and Applied Mathematics at the prestigious ETH in Zürich, Switzerland.

I am excited about this opportunity to apply my unique problem solving skills and analytical abilities to financial markets and trading. The facets of mathematics that have interested me most both during my undergraduate years as well as more recently have always been tightly linked to the efficient solution of real-world tasks by way of mathematics. This was only cemented further last summer, when I worked as a Quantitative Research Intern in FX at the swedish bank Skandinaviska Enskilda Banken (SEB), as well as the summer before which I spent as a visiting researcher at Caltech in Pasadena, California, working on efficient algorithms for real-time planning and decision making in the context of robotics, as well as novel methods in reinforcement learning. What struck me was that many of the key modelling features of a robotics decision problem were similar to those encountered at my previous position where I worked as a summer intern and subsequently part-time with Quantitative Development R&D at the Stockholm-based prop trading firm Ampfield, where I was working on futures-portfolio valuation algorithms utilizing High-Performance Computing techniques and GPU programming. In particular what fascinated me is that while both a robotics path planning task and the optimal trading of futures portfolios may be modelled by the same type of sequential decision process, the difference in problem parameters, problem scale and real-world constraints mean that strikingly different solution approaches are appropriate.

My prior professional experience, as well as my background and continued interest in competitive programming, has given me valuable insight into how technology, research and mathematics may be combined and applied in order to solve complex problems efficiently and analytically. It has also given me broad programming skills in languages such as Python, C++, SQL and Java. In addition, it has piqued my interest in quantitative finance and financial markets, two topics I am very excited to continue developing my expertise in. During previous projects I have gained knowledge about efficient problem solving under pressure, both as part of a team and by myself, for instance during my project "Diagnosing Pneumothorax in X-Ray Images Using Deep Learning" where a colleague and I developed a deep learning network to segment medical x-ray images to detect pneumothorax and alleviate the triage process for medical doctors. In another project, I implemented numerical methods for two-dimensional fluid flow problems in complex geometries. In November 2021, my team and I won the EY Data Case Challenge, where unstructured data were to be analyzed using data-scientific methods. The results were to be presented for a jury. The year prior, my team and I achieved a third place in the programming competition Scania Student Challenge, where the final task consisted of constructing algorithms for autonomous routing of toy truck convoys. Lately, I have been developing my skills chiefly by continued participation in programming contests such as NCPC, and by graduate-level courses in topics that have interested me recently, such as Martingales and Stochastic Processes, Probability Theory, Nonlinear Dynamics, Partial Differential Equations, and Nonparametric Estimation and Regression.

As a person, I am precise, curious, extroverted and driven. In my spare time, I like to exercise, program and read. Ambition, drive and cooperation are of prime importance for me, and I look forward to applying these principles as a Trader at Flow Traders. I look forward to sharing more about my skills and my motivation during an interview.

Best,

Felix Steinberger Eriksson